MING-FENG HO (何銘峰)

jibancat.github.io | mfho@umich.edu | Google Scholar | orcid.org/0000-0002-4457-890X
| 450 Church Street Ann Arbor, MI 48109-1040

CURRENT POSITION

· Postdoctoral Fellow in the Leinweber Center for Theoretical Physics, University of Michigan *2024 Sept -*
Probabilistic modeling for cosmological surveys: DESI, LSST
Mentor: Camille Avestruz

EDUCATION

· Ph.D. student in Physics & Astronomy, University of California, Riverside *2018 - 2024 Sept*
Research advisor: Prof. Simeon Bird
Thesis: Model-driven cosmology with Bayesian machine learning and population inference
· M.S. in Astrophysics, National Taiwan University *2016 - 2018*
· B.S. in Physics, National Taiwan University *2010 - 2014*

RESEARCH INTERESTS

· **Bayesian statistics**: Probabilistic graphical models (PGMs) , hierarchical Bayesian modeling.
· **Uncertainty quantification**: Multi-fidelity methods, emulation, Gaussian processes (GPs).
· **Cosmology**: Lyman-α forest, N-body (hydro) simulations.
· **Black holes**: gravitational waves, primordial black holes (PBHs).

AWARDS AND GRANTS

1. Robert Poe (浦大邦) Memorial Scholarship, UC Riverside *2024*
2. NASA FINESST Fellowship *2021 - 2024*
3. Anne Kernan Award (Outstanding Senior Graduate Student Researcher), UC Riverside *2023*
4. Provost's Scholars Fellowship (honored), UC Riverside *2021*
5. Benjamin C. Shen Award (Outstanding Junior Graduate Student Researcher), UC Riverside *2021*
6. GSA Conference Travel Award, UC Riverside *2019*
7. Student Thesis Award, Physics Society of Taiwan *2019*
8. Dean's Fellowship, UC Riverside *2018*
9. Laureate for Philosophical Treatise (哲學桂冠), National Taiwan University *2012*

RESEARCH MENTORSHIP

Advisor for UCR/UCLA undergraduates advised by Prof. Simeon Bird

· Kevin Hong (co-advised with Sum Qezlou): Animating the evolution of universe using `Blender` [Video] *2022*
· Ryan Tsaio: Automated detection of Lyman limit systems *2023*

Research advisor for King's high school students

· Emma Shah (now UC Berkeley; co-advised with Phoebe R. Upton Sanderbeck): DLA spatial separation on the detection pipeline *2020 - 2021*

· Emma Shah and Rafael Rosales: A better prior for QSO redshift estimation (jibanCat/gpy_dla_detection, Gold medal in the county-level science fair) *2021 - 2022*

Team Lead for Data Science Challenge, Lawrence Livermore National Laboratory

· Adhith Karthikeyan, Alex Chen, Matthew Lee: Deep learning for galaxy/asteroid *2021*

PUBLICATION

Summary

· 20 publications (224 total citations; Google scholar 04/29/2025).
· 5 first-author publications (107 total citations).

First-authored papers

· **Ming-Feng Ho**, Scott Ellis Perkins, Simeon Bird, William Dawson, Nathan Golovich, Jessica R. Lu, Peter McGill, *Investigating the mixing between two black hole populations in LIGO-Virgo-KAGRA GWTC-3,* Phys. Rev. D 110, 063031, September 2024 [2408.09024].
· **Ming-Feng Ho**, Simeon Bird, Martin A. Fernandez, Christian R. Shelton, *MF-Box: Multi-fidelity and multi-scale emulation for the matter power spectrum,* MNRAS, 526(2):2903–2919, December 2023 [2306.03144].
· **Ming-Feng Ho**, Simeon Bird, Christian R. Shelton, *Multi-Fidelity Emulation for the Matter Power Spectrum using Gaussian Processes,* MNRAS, 509(2):2551–2565, January 2022 [2105.01081].
· **Ming-Feng Ho**, Simeon Bird, and Roman Garnett, *Damped Lyman-alpha Absorbers from Sloan Digital Sky Survey DR16Q with Gaussian processes,* MNRAS, 507(1):704–719, October 2021 [2103.10964].
· **Ming-Feng Ho**, Simeon Bird, and Roman Garnett, *Detecting multiple DLAs per spectrum in SDSS DR12 with Gaussian processes,* MNRAS, 496(4):5436–5454, August 2020 [2003.11036].

SCIENTIFIC PRESENTATIONS

Summary

· 20 talks, posters and presentations since 2019
· 6 remote talks, 8 in-person seminar talks, 1 poster, 5 conference talks

Selected Talks

· *Invited Talk: OSU CCAPP Seminar (topic: DESI GP Damped Lyman alpha absorbers)* *Mar 2025*
· *Invited Talk: CfA Seminar (topic: Multi-fidelity emulators and cosmological constraints)* *Nov 2023*
· *Invited Talk: IPMU APEC seminar (topic: PRIYA emulator using Astrid simulations)* *Oct 2023*
· *Invited Talk: KICP seminar (topic: MF emulator)* *Jan 2023*

SERVICE

Selected service

· Referee for Physical Review D, Astrophysical Journal, and MNRAS *2021 - present*
· P&A PeER Mentorship (PAPER) Leader *2023 - present*
· P&A Student Seminar (PASS) Founder *2022 - 2023*
· Head Steward (Riverside), UAW 2865 *2021 - 2022*
· UC Astronomy Osterbrock Sierra Conference Co-I *2021*

PUBLIC OUTREACH

1. UCR's Stargazing Series Presenter and Moderator (× many times) [Video] *2020 - 2021*

2. UCR's Mercury Transit Presenter (hands-on demos) *2019*

WORKING EXPERIENCE

Research Assistant for Digital Humanities Taipei, Taiwan

· Text mining for digital humanities, working at Academia Sinica, Chinese literature and Philosophy *2018*

Magazine Editor/Freelance Writer Taipei, Taiwan

· Editing for Little Newton, and also writing literature, receiving the nation-wide Hakka literature award *2015*

Taiwanese Military Service Keelung, Taiwan

· Digitalize historical court records *2014 - 2015*

OTHER SKILLS

· Languages: Mandarin (native), English, Japanese (limited listening/reading), Taiwanese (limited listening)